Basic Probability

Definition.

- Convergence in distribution: $X_n \xrightarrow{\mathscr{D}} X$ if their cdfs converge pointwise to the cdf of X.
- Convergence in probability: $X_n \xrightarrow{\text{p}} X$ if $\lim \mathbb{P}(|X_n - X| \geq \epsilon) = 0$ for each $\epsilon > 0$.
- Stochastic dominance: $Y \succeq X$ if for each t, $\mathbb{P}(X > t) \leq \mathbb{P}(Y > t)$.

Proposition. *Let F be the CDF of X. If F is continuous and strictly increasing, then $Y \coloneqq F(X) \sim \text{Unif}[0,1]$. If $U \sim \text{Unif}[0,1]$, then $F^{-1}(U) \sim F$.*

Proposition. *$Y \coloneqq h(X)$. If h is injective and differentiable, then*

$$f_Y(y) = f_X(h^{-1}(y)) \cdot \left| \frac{\mathrm{d}h^{-1}(y)}{\mathrm{d}y} \right|.$$

Properties of $\mathbb{E}$, Var, and Cov:

- If $\int_{\mathbb{R}} |x| f(x)\,\mathrm{d}x < \infty$ then $\mathbb{E}(X) = \int_{\mathbb{R}} x f(x)\,\mathrm{d}x$.
- $\mathbb{E}$ commutes with closed (in particular continuous) linear operators.
- If X is stochastically dominated by Y then $\mathbb{E}(X) \leq \mathbb{E}(Y)$.
- $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$.
- $\text{Var}(a + bX) = b^2 \text{Var}(X)$.
- $\text{Cov}(a + bX, c + dY) = bd\,\text{Cov}(X, Y)$.
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\,\text{Cov}(X, Y)$.

If X and Y are independent:

- $\mathbb{E}(XY) = \mathbb{E}(X)\,\mathbb{E}(Y)$, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.
- $\mathbb{E}(X|Y) = \mathbb{E}(X)$ wp 1.
- X and any function of Y are independent.
- $\text{Cov}(X, Y) = 0$. The converse is not true: if $Z \sim N(0,1)$, and S and T are random signs (1 or -1), then $\text{Cov}(SZ, TZ) = 0$.

Proposition.

(i) Law of total expectation / Tower law: $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$.

(ii) We have the following wp 1:

$$\mathbb{E}[g(X)h(Y)|Y] = h(Y)\,\mathbb{E}(g(X)|Y), \mathbb{E}[X|T(Y)] = \mathbb{E}[\mathbb{E}[X|T(Y)]|Y].$$

(iii) Law of total variations: we have

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X)] + \text{Var}[\mathbb{E}(Y|X)],$$

where $\text{Var}(Y|X) \coloneqq \mathbb{E}(Y^2|X) - (\mathbb{E}(Y|X))^2$.

Proposition. *If $X \geq 0$ and there exists an at most countable subset $S = \{x_1, x_2, \dots\}$ of isolated points such that F_X is C^1 on $[0, \infty) \setminus S$, then*

$$\mathbb{E}(X) = \sum_{x \in S} x\mathbb{P}(X = x) + \int_0^\infty x F_X'(x)\,\mathrm{d}x.$$

Joint Distribution

Definition (Jacobian). $Jg(y) \coloneqq \det\left[\partial g_i / \partial y_j\right]$.

Definition (Conditional density). $f_{X|Y}(x|y) \coloneqq f_{X,Y}(x,y)/f_Y(y)$.

Proposition. *If $X : \Omega \to H \subset \mathbb{R}^n$ and $h : H \to G \subset \mathbb{R}^n$, where H and G are open, are such that h is injective and differentiable and $h^{-1} : G \to H$ is differentiable. Then $Y \coloneqq h(X)$ has density*

$$f_Y(y) = \begin{cases} f_X(h^{-1}(y)) \cdot \left|Jh^{-1}(y)\right|, & y \in G \\ 0, & y \notin G. \end{cases}$$

Probability Inequalities

Theorem (Markov). *If $X \geq 0$ and $c > 0$, then $\mathbb{P}(X \geq c) \leq \mathbb{E}(X)/c$. Equality is attained when $\mathbb{P}(X = 0$ or $X = c) = 1$.*

Theorem (Chebychev). *If $c > 0$, then*

$$\mathbb{P}(|X - \mu| \geq c) \leq \frac{\mathbb{E}[(X-\mu)^2]}{c^2}, \quad \forall \mu.$$

Theorem (Chernoff). *If $c \in \mathbb{R}$ and $t > 0$, then*

$$\mathbb{P}(X \geq c) \leq e^{-tc}\,\mathbb{E}(e^{tX}), \quad \mathbb{P}(X \leq c) \leq e^{tc}\,\mathbb{E}(e^{-tX}).$$

Theorem (Jensen). *If $f : \mathbb{R} \to \mathbb{R}$ is convex and X is an rv with finite expectation, then $f(\mathbb{E}\,X) \leq \mathbb{E}\,f(X)$.*

Applications of Jensen:

- If f is concave, then $f(\mathbb{E}\,X) \geq \mathbb{E}\,f(X)$.
- $(\mathbb{E}\,X)^2 \leq \mathbb{E}\,X^2$ and $\log \mathbb{E}\,X \geq \mathbb{E} \log X$.
- If $x_1, x_2, \dots, x_n > 0$, $p_i \geq 0$, and $\sum p_i = 1$, then $\prod x_i^{p_i} \leq \sum p_i x_i$. When $p_i = 1/n$, this is AM-GM.
- **Hölder's inequality**: If $p, q > 0$ and $1/p + 1/q = 1$, then $\mathbb{E}(XY) \leq (\mathbb{E}\,X^p)^{1/p} \cdot (\mathbb{E}\,Y^q)^{1/q}$.
- **Cauchy Inequality**: $\mathbb{E}|XY| \leq \sqrt{\mathbb{E}\,X^2}\sqrt{\mathbb{E}\,Y^2}$.

Asymptotic Behavior

Theorem (WLLN). *Let $X_1, X_2, \dots$ be iid with finite expectation μ and variance σ^2. Then as $n \to \infty$, $\overline{X}_n \xrightarrow{\text{p}} \mu$.*

Theorem (CLT). *If $X_1, X_2, \dots$ are iid, $\mathbb{E}(X_i) = \mu$, and $\text{Var}(X_i) = \sigma^2$, then $\sqrt{n}(\overline{X}_n - \mu) \xrightarrow{\mathscr{D}} \mathcal{N}(0, \sigma^2)$.*

Proposition (Large Deviations). *Let $X_1, X_2, \dots$ be iid with finite expectation μ and variance σ^2. Let $c > \mu$. Then*

$$\lim_{n \to \infty} \frac{1}{n} \log \mathbb{P}(\overline{X}_n > c) = -\sup_t [tc - \kappa(t)],$$

where $\kappa(t) = \log \mathbb{E}(e^{tX})$.

$\mathbb{E}[e^{tX}]$ is the moment generating function, $\kappa(t)$ is the cumulant generating function, and $\sup_t [tc - \kappa(t)]$ is the Legendre transform.

Moment Generating Functions

$M_X : \mathbb{R} \ni t \mapsto \mathbb{E}[e^{tX}] \in [0, \infty]$.
Properties:

- $\mathbb{E}[X^n] = M_X^{(n)}(0)$ when Fubini grants so. $\mathbb{E}[e^{tX}] = \sum_{n=0}^\infty \frac{t^n\,\mathbb{E}(X^n)}{n!}$.
- $M_{cX}(t) = M_X(ct)$.
- If X and Y are independent, then $M_{X+Y}(t) = M_X(t) + M_Y(t)$.
- If X_i are iid, then $M_{\sum X_i} = \prod M_{X_i}$.
- $X_n \xrightarrow{\mathscr{D}} X$ if and only if $M_{X_n} \to M_X$ in a neighborhood of 0.

Point Estimates

Bias: $\mathbb{E}(\hat{\lambda}) - \lambda$. Variance: $\text{Var}[\hat{\lambda}]$. Mean Squared Error: $\text{MSE}[\hat{\lambda}] = \mathbb{E}[(\hat{\lambda} - \lambda)^2] = \text{Bias}^2 + \text{Variance}$.

Definition (Consistency). $\hat{\theta}_n \xrightarrow{\text{p}} \theta$.

Example. Modeling lifetime $T : \Omega \to [0, \infty)$.

- The **survival** function: $S : [0, \infty) \ni x \longmapsto \mathbb{P}(T > x) = 1 - F_Y(x) \in [0, 1]$.
- The **failure rate** function: $h(x) \coloneqq f(x)/S(x)$. Note that $\mathbb{P}(T \leq x + \Delta x | T > x) \approx \Delta x \cdot h(x)$.

Given h we can recover f: $f(x) = h(x)\exp\left(-\int_0^x h\right)$. If $h(x) = \lambda$ is constant, then $T \sim \mathrm{Exp}(\lambda)$.

Method of Moments.

Theorem (Continuous Mapping Theorem)**.** *Let g be continuous. Then $X_n \xrightarrow{\mathrm{p}} X \implies g(X_n) \xrightarrow{\mathrm{p}} g(X)$ and $X_n \xrightarrow{\mathscr{D}} X \implies g(X_n) \xrightarrow{\mathscr{D}} g(X)$.*

Lemma (Slutsky)**.** *If $X_n \xrightarrow{\mathscr{D}} X$ and $Y_n \xrightarrow{\mathrm{p}} c$, where c is a constant, then $X_n + Y_n \xrightarrow{\mathscr{D}} X + c$ and $X_nY_n \xrightarrow{\mathscr{D}} cX$.*

Theorem (Delta Method)**.** *If $\sqrt{n}(X_n - \theta) \xrightarrow{\mathscr{D}} \mathcal{N}(0, \sigma^2)$ and $g \in C^1$, then $\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{\mathscr{D}} \mathcal{N}(0, \sigma^2[g'(\theta)]^2)$.*

Properties of MoM estimators:

- Consistency by WLLN and the continuous mapping theorem.
- Asymptotic normality by CLT and Delta Method.

Likelihood Theory.

Definition (Score function)**.**

$$\dot{\ell}_n(\theta) \coloneqq \frac{\partial}{\partial\theta}\ell_n(\theta) = \sum_{i=1}^n \frac{\frac{\partial f}{\partial\theta}(x_i,\theta)}{f(x_i,\theta)} = \sum_{i=1}^n \frac{f'(x_i,\theta)}{f(x_i,\theta)}.$$

Proposition. *If $f(x,\theta)$ has common support, then $\mathbb{E}_{\theta_0}\left[L_n(\theta)/L_n(\theta_0)\right] = 1$. Equivalently, $\mathbb{E}\left[\exp\left(\ell_n(\theta) - \ell_n(\theta_0)\right)\right] = 1$.*

Proposition. *If the density functions are smooth, then $\mathbb{E}_\theta\left[\dot{\ell}_n(\theta)\right] = 0$ and $-\mathbb{E}_\theta\left[\ddot{\ell}_n(\theta)\right] = \mathbb{E}[\dot{\ell}_n(\theta)^2]$.*

Definition (Fisher Information)**.** $I(\theta) \coloneqq \mathbb{E}_\theta[\dot{\ell}(\theta)^2] = \mathbb{E}_\theta[-\ddot{\ell}(\theta)]$. That is, $I(\theta) \coloneqq \mathbb{E}\left[\left(\frac{\partial}{\partial\theta}\log f(X,\theta)\right)^2\right] = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2}\log f(X,\theta)\right]$. Expectation is taken wrt $X \sim f(x,\theta)$.

Theorem (Cramér–Rao Inequality)**.** *Let $T(X_n)$ be an unbiased estimator for $g(\theta)$. Then, $\mathrm{Var}[T(X_n)] \geq [g'(\theta)]^2/nI(\theta)$.*
Equality is attained if and only if $\mathrm{Corr}(\hat{\theta}(X), \dot{\ell}(X)) = 1$, i.e., if $\hat{\theta}(X)$ and $\dot{\ell}(X)$ are linearly related rvs. That is, $\dot{\ell}(\theta) = \alpha(\theta)\hat{\theta}(X) + \beta(\theta)$ for functions α and β that do not depend on X.

Proposition. *There exists an unbiased estimator $\hat{\theta}(X)$ of θ whose variance attains the Cramér–Rao lower bound if and only if the score can be expressed in the form $\dot{\ell}(\theta) = I(\theta)\left\{\hat{\theta}(X) - \theta\right\}$ or, equivalently, if and only if the function $\frac{\dot{\ell}(\theta)}{I(\theta)} + \theta$ does not depend on θ.*

Theorem (Fisher)**.** *$X_1, \ldots, X_n \overset{\mathrm{iid}}{\sim} f(x|\theta_0)$. As $n \to \infty$, we have*

$$\sqrt{nI(\theta_0)}\cdot(\hat{\theta} - \theta_0) \xrightarrow{\mathscr{D}} \mathcal{N}(0,1) \quad \textit{and} \quad \sqrt{nI(\hat{\theta})}\cdot(\hat{\theta} - \theta_0) \xrightarrow{\mathscr{D}} \mathcal{N}(0,1)$$

Think also $\hat{\theta} \approx \mathcal{N}\left(\theta_0, 1/nI(\theta_0)\right)$.

Proposition. *Assumptions: densities are smooth, have common support, and are distinct for different parameters. Properties of MLE under the above assumptions: consistent, asymptotically normal, known and optimal asymptotic variance (**efficiency**), invariance (if $\hat{\theta}_n$ is an MLE of θ, then $\hat{\tau}_n \coloneqq g(\hat{\theta}_n)$ is an MLE of $g(\theta)$).*

Confidence Intervals

If $X_i \overset{\mathrm{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, n small: Recall exact distribution results.
CI for μ: $\left[\overline{X} + \frac{t_{L,n-1}S}{\sqrt{n}}, \overline{X} + \frac{t_{R,n-1}S}{\sqrt{n}}\right]$. CI for σ^2: $\left[\frac{(n-1)S^2}{\chi^2_{R,n-1}}, \frac{(n-1)S^2}{\chi^2_{L,n-1}}\right]$.
If $X_i \overset{\mathrm{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, n large: use CLT.
CI for μ: $\left[\overline{X} - \frac{z_R\sigma}{\sqrt{n}}, \overline{X} - \frac{z_L\sigma}{\sqrt{n}}\right]$ (use S to approximate σ if needed).
If $\hat{\theta}$ is MLE: use Fisher.
CI for θ: $\left[\hat{\theta} - \frac{z_R}{\sqrt{nI(\hat{\theta})}}, \hat{\theta} - \frac{z_L}{\sqrt{nI(\hat{\theta})}}\right]$.

The Bootstrap

Definition (Empirical Dsitribution Function)**.** Let $X_i \overset{\mathrm{iid}}{\sim} F$. The **edf**, $\hat{F}_n$, is the cdf that puts probability $1/n$ at each X_i. $\hat{F}_n(x) \coloneqq \frac{1}{n}\sum \mathbb{1}_{\{X_i \leq x\}}$.

Properties of the edf:

- $\mathbb{E}[\hat{F}(x)] = F(x)$.
- $\mathrm{Var}(\hat{F}(x)) = F(x)(1 - F(x))/n$.
- By WLLN, $\hat{F}(x) \xrightarrow{\mathrm{p}} F(x)$.
- $\hat{F}_n(x) \to F(x)$ uniformly. $\sup_x\left|\hat{F}_n(x) - F(x)\right| \to 0$.

Distributions and Properties

$X \sim$	$\mathrm{supp}\,\mathbb{P}(X = \cdot)$	$\mathbb{P}(X = x)$	$\mathbb{E}$	Var
$\mathrm{Binom}(n,p)$	$\{0, 1, \ldots, n\}$	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$
$\mathrm{Geom}(p)$	$\mathbb{N}$	$(1-p)^{x-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$\mathrm{Poisson}(\lambda)$	$\mathbb{N} \cup \{0\}$	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ

$X \sim$	$\mathrm{supp}\, f$	f	$\mathbb{E}$	Var
$\mathrm{Unif}(a,b)$	$[a,b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$\mathcal{N}(\mu,\sigma^2)$	$(-\infty,\infty)$	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
$\mathrm{Exp}(\lambda)$	$[0,\infty)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
$\mathrm{Gamma}(\alpha,\beta)$	$(0,\infty)$	$\frac{\beta^\alpha x^{\alpha-1}e^{-\beta x}}{\Gamma(\alpha)}$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
$\mathrm{Beta}(\alpha,\beta)$	$(0,1)$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

Properties of $\mathcal{N}$: Let $X_i \overset{\mathrm{iid}}{\sim} \mathcal{N}(\mu,\sigma^2)$. Let $S^2 \coloneqq \frac{1}{n-1}\sum(X_i - \overline{X})^2$.

- $\mathbb{E}[S^2] = \sigma^2$.
- $\overline{X}$ and S^2 are independent.
- $\sqrt{n}\cdot\frac{\overline{X}-\mu}{S} \sim t_{n-1}$ and $(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$.

Properties of Exp:

(i) "Memoryless": $\mathbb{P}(T \leq x + y | T > x) = \mathbb{P}(T \leq y)$.
(ii) $\mathrm{Exp}(\lambda) = \mathrm{Gamma}(1,\lambda)$.

Properties of Gamma:

(i) If $X_i \overset{\mathrm{iid}}{\sim} \mathrm{Gamma}(\alpha_i,\beta)$, then $\sum X_i \sim \mathrm{Gamma}\left(\sum\alpha_i,\beta\right)$.
(ii) If $X \sim \mathrm{Gamma}(\alpha,\beta)$ and $\alpha > 1$, then $\mathbb{E}\left[1/X\right] = \frac{\beta}{\alpha-1}$.
(iii) If $X \sim \mathrm{Gamma}(\alpha,\beta)$, then $\beta X \sim \mathrm{Gamma}\left(\alpha,1\right)$.

Miscellaneous

Definition. $\Gamma(\lambda) \coloneqq \int_0^\infty e^{-x}x^{\lambda-1}\,\mathrm{d}x$.

Properties: $\Gamma(1) = 1$; $\Gamma(1/2) = \sqrt{\pi}$; $\Gamma(x+1) = x\Gamma(x)$; $\Gamma(n) = (n-1)!, \forall n \in \mathbb{N}$.